Opinion : Cold atomic ensembles for light-matter quantum technologies, my vision for the road ahead

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Disclaimer : this (debatable) opinion is focused on a cold atom point of view. I do not discuss other light-matter platforms here, and for now I think it is too early to say if any platform is better than another: they all have their own strengths and weaknesses, and many systems look promising.

My takes for the road ahead

1. **If the rise of cold atoms for quantum computing continues, quantum repeaters and routers based on cold atoms will become increasingly relevant.**

   Why do I say that cold atom quantum computing is rising fast : In 2024, a group at Caltech developed a cold atom machine with around 6k qubits [1], it's quite a lot already. End of 2025, Lukin's group demonstrated the first error corrected platform with cold atoms [2]. Also, there is a path for scaling up to hundreds of thousands of qubits: using metasurfaces for tweezer generation (instead of SLM's), as demonstrated in this recent paper [3] (january 2026) where they show a tweezer array of 360000 traps.

2. **Sources of light-matter (or photon-photon) entanglement should not be stochastic (such as DLCZ, four wave mixing or SPDC). Deterministic sources of entanglement hold much more potential**

   In stochastic protocols such as DLCZ, FWM or SPDC, the success rate of entanglement generation is given by a probability $p \ll 1$, that needs to be kept low to keep a high fidelity due to multi excitations degrading the state. Deterministic sources remove that intrinsic trade-off: in the ideal case, one entangled pair is produced on demand each cycle, so the rate is only set by the total efficiency. These can be done with single-atom in cavities (see, for example [4, 5]), or with Rydberg superatoms in cavities [6–9].

3. **The community needs to develop non-demolition measurements to check if, after storage, a qubit has been correctly loaded in the memory or not, without impacting its storage**.

   I would call these "error-detected" quantum memories. Such a capability would exponentially increase the success rates of entanglement distribution. Such techniques need a non-linearity. A few experimental realizations have been achieved [6, 10, 11], and generally one needs a cavity for this to work.

4. **The protocols implemented need to be close to 100% efficiency, and have long coherence times**.

   The former is achievable by having a cavity around a collection of atoms [12–14] (I think this is preferable to having large ensembles [15–17], or high density ensembles). For the latter, the difficulty is to mitigate the dephasing due to atomic motion. This can be achieved using lattices [18–20], or using optical techniques to supress this motional dephasing, where nice new techniques have been developed these past 3 years [21–23].

5. **These quantum memories should have the ability to perform in-memory two qubits gates between stored qubits**.

   This is equivalent to performing quantum gates between flying photonic qubits. For example, this would allow to perform Bell-state measurements above the 50% limit given by linear optics (this is probably a necessity). For now, breaking the 50% limit has not been demonstrated experimentally this way, but I believe it will be tackled in the next few years. Since 2020, such two qubit gates have been realized experimentally with Rydberg blockaded ensembles [10, 24, 25]. Single qubit gates in memory have also been demonstrated [7, 26], showing that all the ingredients are currently available to perform in-memory Bell-state measurements. Memory efficiency needs to be improved since it constitutes an upper bound for the success probability of this measurement.

6. **Routing photonic qubits in a network, conditionally (based on the state of other photonic qubits), and without detroying the qubits, can be useful**.

   It seems rather obvious that a quantum network node should be able to "route" photons, by this I means change their direction, switch on-off their propagation, reshape them and other similar type of operations. Again, small atomic ensembles coupled to rydberg state are able to do such switching operations, see for example [27, 28]. For rerouting photons, one can use the properties of phase matching in spinwaves to change the direction of reemission after storage (see [29] for an example). Shaping or reshaping can be easily done

with modulating the control pulse (see [30] for example). Conditionality can be done using non-demolition measurements in the first place (see take number 4).

7. **We can already build a single machine that can do everything needed for a fonctional network: generate light-matter entanglement, store photonic qubits, perform gates and measurements on them, do non-demolition detections of photonic qubits, everything deterministically**.

   I think all of the above could be done with a small atomic ensemble in a cavity. Travelling photonic qubits would be encoded in temporal bins, efficient storage could be done with EIT in the spin states, using Raman beams to transfer the spin wave to other Zeeman state in between temporal modes (qubits could be written in spin waves of different Zeeman states for example). When two qubits are in the memory, gates can be performed using Rydberg blockade, assuming the whole ensemble fits inside the blockade radius. Non-demolition measurements can be done using Rydberg-EIT and the cavity reflection properties. Light-matter entanglement can also be generated deterministically from such system using Rydberg blockade. Could also be implemented using single atoms coupled to cavities, though it seems generally harder to achieve perfect efficiencies (weaker coupling and no collective enhancement has to be compensated with insane cavities).

8. **Of course, there are still quite a lot of challenges to overcome, and I am not sure if they will ever be**.

   Some are technical, but some are inherent to this system, one of them being the scalability (cold atoms systems are hard to miniaturize). Also it's hard for me to see how to scale up the multiplexing capabilities of one device while keeping the capabilities listed above. An idea to increase the multiplexing capabilities of cold atoms is to use acousto optic deflectors to adress different part of the cloud [31], though for me it is still unclear how to keep high efficiency of operations with this technique. Perhaps one way is to use a cavity that can support many spatial modes, inspired from this impressive work [32]. Another topic is the telecom compatibility, I think we should study more how to use the available transitions in the C-band and O-band that many alkaline have [33–35], because if combined with rydberg blockade, one could theoretically generate entangled pairs deterministically, with one photon in the telecom, without multiphoton excitation, yet this is left to be demonstrated. Also, telecom compatibility is not necessary for short distance scale quantum networks.

9. **Other applications of this technology are also worth to investigate**.

   The principal one I see is highly efficient and low noise quantum transduction. Especially from optical to microwave, it's a difficult task when one wants to do it efficiently and without noise. It turns out that cold atomic ensembles are an amazing tool for this ! This recent preprint from february 2026 [36] shows a microwave to optical transduction with 90% efficiency, this paper is really amazing to me. Otherwise, the group of Jonathan Simon demonstrated a 60% efficiency transduction using a Rydberg polariton coupled to both an optical and microwave cavity [37]. Another interesting topic is to use the atomic ensemble as a linear quantum processor: spin waves are basically bosonic, and some operations on these collective states are equivalent to beamsplitter operations or phase shifts. Thus, some people are thinking about implementing the equivalent of linear programmable optical quantum circuits, but all in-memory. The idea is well explained in this nice paper [38], and recently, quite a lot of experiments have been investigating the coherent manipulation of collective states [26, 39–42].

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